Chapter 11: Architecture Domains

Study Guide – based on Mastering the TOGAF® Standard by Eric Jager (2025), Chapter 11 (pp. 279-294)

Overview

Chapter 11 (pp. 279-294) of *Mastering the TOGAF® Standard* by Eric Jager explains the **four core architecture domains** (Business, Data, Application, Technology): where they came from, how they differ from architecture **layers**, which **cross-cutting concerns** run through them, and how the roles from Chapter 10 map onto them. It completes a chain across three chapters. Chapter 9 decides how the capability is positioned, Chapter 10 decides which roles that needs, and Chapter 11 decides which domains each role covers. It is part of the PINK and GREEN reading tours.

Detailed Section Breakdown

Introduction (p. 279)

The core roles relate to the capability's positioning. The Chief/Enterprise Architect supports strategy. The Segment Architect supports portfolios and, sometimes, large projects. The Solution Architect supports solution delivery. Whatever the role, an architect works with one or more domains.

11.1 Four Architecture Domains (pp. 280-286)

Domain	TOGAF description (Figure 11-1, p. 280)
Business Architecture	Defines the business strategy, governance, organization, and key business capabilities (or processes, per the summary), so that IT supports business objectives.
Data Architecture	Describes the structure of logical and physical data assets and data management resources. Enables data governance and informed decisions.
Application Architecture	Blueprint for the individual systems to be deployed, their interactions, and their relationships to core business processes.
Technology Architecture	Details the hardware, software, and network infrastructure required to support mission-critical applications.

Data and Application Architecture together are called **Information Systems Architectures**, which is Phase C of the ADM (p. 281).

History of the domains (pp. 281-282)

- **TOGAF 7** recognized only Technology Architecture.
- **TOGAF 8 (2002)**: Business Architecture; Information Systems Architectures (“Data **or Information** Architecture” plus Application Architecture); Technology Architecture. The words “or Information” later disappeared.
- The four-domain classification has existed “since the eighth edition.” (The book later says “the 2008 eighth edition” on p. 285, which conflicts with 2002 on p. 282. Treat **TOGAF 8 = 2002** as the reliable date.)
- With closer alignment to the **BIZBOK Guide**, which emphasizes information, Information Architecture may return.

Alternative models: the Interoperability Model (pp. 282-283)

The **Interoperability Model** (five-layer model, Figure 11-2, p. 283) is common in healthcare, particularly in the Netherlands (Nictiz). It splits Business Architecture into **organization** and **processes**, and uses **Information** instead of Data. It also has two preconditional columns that apply to all layers: legislation and regulations, and security. Otherwise it essentially matches TOGAF's four domains.

Data versus information (pp. 283-285)

- TOGAF's Data Architecture uses the **data entity** (Ch. 5). The BIZBOK Guide uses the more detailed **information concept**. They are not directly comparable.
- **Data + Context = Information.** Examples: “1 Primary Lane” means little until it is labelled the President's home address. “22, 24, 26, 21, 23” becomes information once it is labelled next week's expected temperatures (p. 284).

Completeness (pp. 285-286)

- TOGAF: *a complete Enterprise Architecture should address all four architecture domains* (p. 285).
- Time and resource limits often prevent this. Architecture descriptions are purpose-driven, and defining the questions they answer is part of the initial ADM phase.
- Example: if only technology options are being assessed and business processes are not changing, a full Business Architecture may be unnecessary. Data, Application, and Technology still build on the Business Architecture, so it must be understood.
- An architecture without all four domains is, by definition, **not a complete EA**. The risks are inconsistency and integration problems, which add cost later. The architect must explain these trade-offs to management.

11.2 Difference in Domains and Layers (pp. 286-289)

EA was technology-focused around 1990 but now covers all aspects of the organization. Organizations must contribute too, for example by defining strategy, and EA helps shape drivers, goals, objectives, and initiatives (p. 286). Domains and layers are often confused, but they are different:

Aspect	Architecture layers	Architecture domains
What	Levels of abstraction within a **system or Solution Architecture**; logical divisions of software by functionality	Broad divisions of the overall architecture; **areas of concern or expertise**
Examples	Presentation, business logic, data access, infrastructure (Figure 11-3, p. 287)	Business, Data, Application, Technology (+ cross-cutting concerns)
Purpose	Modularity, maintainability, reusability; layers talk through predefined interfaces	Categorize concerns and perspectives to address during design
Relationship	Organize components within a domain	Contain layers; the two are **complementary**

Cross-cutting concerns (Figure 11-4, p. 289) cut across all domains: **security, motivation, and governance**. Security Architecture is sometimes called a domain but is mostly treated as cross-cutting. Jager notes that the **ACMM** maturity model (Ch. 15) lists IT security as one of its nine elements, but only covers the IT part and ignores security as a cross-cutting concern (p. 288).

11.3 Mapping Roles to Domains (pp. 289-292)

Role	Domain scope	Figure
Chief / Enterprise Architect	The **entire ecosystem**, all four domains. The Chief Architect focuses on strategic issues. Without both roles, the Enterprise Architect does both.	Figure 11-6, p. 290
Segment Architect	Four specializations (Business, Data, Application, Technology Architect), each responsible for one domain.	Figure 11-7, p. 291
Solution Architect	A defined project or a significant part of one, usually technical. In nine out of ten cases this is the Technology domain. Specializations: Cloud, Network, Software Architect.	Figure 11-8, p. 292

Practice may differ, but each role has its own area of interest. Don't hold false expectations of what a role covers (p. 292).

11.4 Summary (pp. 292-294)

The summary repeats the four domains, the Interoperability Model alternative, the completeness rule, domains versus layers, cross-cutting concerns, and the role-to-domain mapping. (The summary's opening line says “four core architecture roles” when it means **domains**, a slip in the book.)

Key Terms & Definitions

Business Architecture (p. 280) — Business strategy, governance, organization, and key business capabilities and processes.

Data Architecture (p. 280) — Structure of logical and physical data assets and data management resources.

Application Architecture (p. 281) — Blueprint of systems, their interactions, and their relationships to business processes.

Technology Architecture (p. 281) — Hardware, software, and network infrastructure supporting mission-critical applications.

Information Systems Architectures (p. 281) — TOGAF's collective name for Data + Application Architecture (ADM Phase C).

Interoperability Model (p. 283) — A five-layer model used in healthcare. Splits Business into organization and processes, and uses Information instead of Data.

Information concept (p. 283) — A BIZBOK Guide concept, more granular than TOGAF's data entity.

Data + Context = Information (p. 284) — Raw data becomes information once context gives it meaning.

Architecture layer (p. 287) — A level of abstraction within a system (presentation, business logic, data access, infrastructure).

Architecture domain (p. 287) — A broad area of concern or expertise addressed by architects.

Cross-cutting concern (p. 288) — A concern spanning all domains: security, motivation, governance.

Figures & Tables Referenced

Item	Page	Description
Figure 11-1 – The four core architecture domains	p. 280	Data and Application are grouped as Information Systems Architectures.

Item	Page	Description
Figure 11-2 – Interoperability Model (five-layer model)	p. 283	Healthcare alternative to the four domains.
Figure 11-3 – Architecture layers	p. 287	Presentation, business logic, data access, infrastructure.
Figure 11-4 – Domains and cross-cutting concerns	p. 289	Security, motivation, and governance across all domains.
Figure 11-5 – Roles/specializations and domains	p. 290	Overall role-to-domain mapping.
Figure 11-6 – Scope of the Chief/Enterprise Architect	p. 290	All four domains.
Figure 11-7 – Scope of the Segment Architect	p. 291	One domain per specialization.
Figure 11-8 – Scope of the Solution Architect	p. 292	Mostly the Technology domain.

Key Takeaways / Exam Focus Points

- The four domains: Business, Data, Application, Technology. **Data + Application = Information Systems Architectures** (Phase C) (pp. 280-281).
- TOGAF 7 had only Technology Architecture. TOGAF 8 (2002) introduced the four-domain structure with “Data or Information” (pp. 281-282).
- A complete EA addresses all four domains. Anything less is, by definition, not a complete EA and risks integration problems (p. 285).
- Layers are not domains. Layers organize a system's components (presentation, business logic, data access, infrastructure). Domains categorize areas of concern (pp. 287-289).
- Cross-cutting concerns: **security, motivation, governance** (p. 288).
- Role scope: Enterprise covers all four domains. Segment covers one domain per specialization. Solution covers mostly Technology (pp. 290-292).
- Data + Context = Information. BIZBOK's information concept is more granular than TOGAF's data entity (pp. 283-284).

Self-Check Questions

Q: Which two domains make up Information Systems Architectures?

A: Data Architecture and Application Architecture (p. 281).

Q: How were the domains classified in TOGAF 7 and TOGAF 8?

A: TOGAF 7 had only Technology Architecture. TOGAF 8 (2002) had Business Architecture, Information Systems Architectures (Data or Information, plus Application), and Technology Architecture (pp. 281-282).

Q: How does the Interoperability Model differ from TOGAF's four domains?

A: It splits Business into organization and processes and uses Information instead of Data. It also adds legislation/regulations and security as preconditional columns (p. 283).

Q: Can an architecture that skips a domain be a complete EA? What are the risks?

A: No, by definition it is not complete. The risks are inconsistency and integration challenges, which add cost and risk later, so the architect must explain the trade-offs (pp. 285-286).

Q: Distinguish architecture layers from architecture domains.

A: Layers are levels of abstraction within a system (presentation, business logic, data access, infrastructure) that communicate through interfaces. Domains are broad areas of concern for the whole architecture. They are complementary (pp. 287-289).

Q: Name the three cross-cutting concerns.

A: Security, motivation, and governance (p. 288).

Q: Which domain does a Solution Architect usually work in, and why?

A: Technology, in nine out of ten cases, because a Solution Architect's scope is usually a technical project (p. 291).

Q: Explain “Data + Context = Information” with an example.

A: The numbers 22, 24, 26, 21, 23 are raw data. Labelled as next week's expected temperatures, they become information (p. 284).